

Scratchwork and Studies related to TFIM

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ABSTRACT: I plan (and wish) to write **three documents** related to this research problem.

- **scratchpad and studying:** on this scratchpad, where I lay the foundation for a detailed solution.
- **detailed solution:** this solution should lay the foundation for the research paper (hopefully) if we do end up writing.
- **Original Draft for Pre-Print.**

First of all we look at the *Transverse Field Ising Model*.

Contents

1	The Transverse Field Ising Model	1
2	Dualities as Isomorphisms: How to use Bond Algebra	4
3	The Toric Code (studied [5])	6
4	Dummy Introduction	9

1 The Transverse Field Ising Model

For now we will work with Spin $\frac{1}{2}$ particles. The spin of the particles are represented by a ket $|\uparrow\rangle = |1\rangle \in \mathcal{H}, |\downarrow\rangle = |0\rangle \in \mathcal{H}$ where $\mathcal{H} = \mathbb{C}^2$ is a Hilbert Space for spin $\frac{1}{2}$.

Let's construct a 1+1d lattice with each site labeled as j ranging from $j = 1$ to $j = L$. Each site j of the lattice is assigned to a Hilbert Space $\mathcal{H}_j = \mathbb{C}^2$. The Hilbert Space for the lattice $\mathcal{H}$ is defined by

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \cdots \otimes \mathcal{H}_L = \bigotimes_{j=1}^L \mathcal{H}_j \quad (1.1)$$

Definition 1: Transverse Field Ising Model

The Hamiltonian for the Transverse Field Ising Model at Critical State on a lattice $H : \mathcal{H} \rightarrow \mathcal{H}$ is defined by

$$H = - \sum_{i=1}^L (Z_{i-1} Z_i + X_i)$$

where Z_i, X_i are Pauli Matrices, in $|1\rangle, |0\rangle$ basis given by:

$$Z_i = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}_{i\text{-th site}} \quad \text{and} \quad X_i = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}_{i\text{-th site}}$$

Periodic boundary condition is invoked through $Z_{L+1}, X_{L+1} = Z_1, X_1$.

Intuitively I like to think this $Z_{j-1}Z_j$ as the “bond” term (between $j - 1$ and j site) and X_j as the site term.

Kramers Wannier Transformation and the Problem

From the Seiberg et al. paper on Non-invertible symmetries [3] using equation 2.12 we know that the Kramers-Wannier Transformation is given by

$$\begin{aligned} X_j &\rightarrow Z_{j-1}Z_j \\ Z_{j-1}Z_j &\rightarrow X_{j-1} \end{aligned} \tag{1.2}$$

This transformation is a automorphism. The original and transformed Hamiltonian have an isomorphism between them, so they are mathematically the same object. We term this a duality.

$$H = - \sum_{i=1}^L (Z_{i-1}Z_i + X_i) \rightarrow - \sum_{i=1}^L (X_{i-1} + Z_{i-1}Z_i)$$

So in my intuition, each “bond” terms become “site” terms and vice-versa. From the same paper, from the analysis after the equation we know that there is no Unitary Operator that can achieve this transformation.

The core problem can be shaped into finding a valid method to transform between these two dualities.

The Non-inv-lsm paper ([3]) solves this problem through the following method

- Extend the Hilbert Space by adding a site (basically extending the lattice with one new site).
- Introduce a Topological Defect (discussed in the paper, on how to move and fuse them).
- Through motion and fusion of the defects, achieve the KW Transformation.

The issue with this process is that we define the topological defect kind of from guesstimates and scratchwork. It’s not necessarily a generalized solid method.

For us, we have a Bond Algebra theorem that leads to a different route of solution. Which I think is quite general and significantly more elegant.

Introductory Discussions on Bond Algebra

Before we start, let me drop the first three important definition and theorem that will be very important to whatever we do later on.

Definition 2: Von Neumann Algebra

Let $\mathcal{H}$ be a Hilbert Space and let $B(\mathcal{H})$ denote the algebra of bounded operators on $\mathcal{H}$. Bounded operator $\mathcal{O}$ means that there exists a finite number $0 \leq C$ such that $\|\mathcal{O}v\| \leq C\|v\|$ for every vector $v \in \mathcal{H}$.

If $S \subset B(\mathcal{H})$ is an arbitrary set, its commutant $S' \in B(\mathcal{H})$ is the sub-algebra defined by

$$S' = \{\mathcal{O} \in B(\mathcal{H}) \mid \forall \mathcal{R} \in S, \mathcal{O}\mathcal{R} = \mathcal{R}\mathcal{O}\}$$

A sub-algebra (as defined above) $\mathcal{A}$ is a Von Neumann Algebra if it satisfies

- $I \in \mathcal{A}$
- $\mathcal{O} \in \mathcal{A} \Rightarrow \mathcal{O}^\dagger \in \mathcal{A}$
- Equal to bycommutant $\mathcal{A} = \mathcal{A}''$

For a set of bonds $\hat{h}_j$, the Von Neumann Algebra $\mathcal{A}\{\hat{h}_j\}$ is

$$\mathcal{A}\{h_j\} = \{\hat{I}, h_j, h_j^\dagger, h_j h_{j'}, h_j^\dagger h_{j'}, h_j h_{j'}^\dagger, h_j^\dagger h_{j'}^\dagger, \dots\}$$

This means if any operator $\mathcal{O} \in \mathcal{A}\{h_j\}$ then $\mathcal{O}^\dagger \in \mathcal{A}\{h_j\}$ also. This sets up with all the Hermitian Qualities for the problem, a favorable algebra.

Definition 3: Bond Algebra Definition

A bond algebra for the Hamiltonian H where $H = \sum_j \lambda_j \hat{h}_j$ with bond decomposition $\{h_j\}_j$, is the *Von Neumann Algebra* $\mathcal{A}\{h_j\}$ generated by the bonds. Here λ_j is a complex-valued coupling constant and $\hat{h}_j$ is a bond operator found after bond decomposition.

First of all note that the j in Definition 3 can be completely general. For this problem this will probably mean the “site index” but for other problems this could be anything: from particles to point x in space of a Fourier Mode. It’s a suitable label for the desired problem.

Bond Decomposition means that the Hamiltonian H could be broken down to different

bond operators.

$$H = \sum_n \lambda_n \hat{h}_n = \sum_{\Gamma} \lambda'_{\Gamma} \hat{h}'_{\Gamma}$$

Various decompositions can lead to different bond algebras.

Theorem 1: insane theorem

Let $\mathcal{A}_i$ be von Neumann algebras of operators on the Hilbert Spaces $\mathcal{H}_i$ for $i = 1, 2$. If $\Phi : \mathcal{A}_1 \rightarrow \mathcal{A}_2$ is an *isomorphism* then

- there exists a Hilbert Space M
- and there exists a Unitary Transformation $U : \mathcal{H}_1 \otimes M \rightarrow \mathcal{H}_2 \otimes M$ such that

$$\Phi(\mathcal{O}) \otimes I = U (\mathcal{O} \otimes 1) U^\dagger$$

$$I \in M$$

$$\mathcal{O} \in \mathcal{H}_1$$

2 Dualities as Isomorphisms: How to use Bond Algebra

Definition 4: Duality as Isomorphism

Two Hamiltonians H_1 and H_2 are called *dual* if there is a bond algebra A_{H_1} for H_1 isomorphic to some bond algebra A_{H_2} for H_2 , and if the isomorphism $\Phi : A_{H_1} \rightarrow A_{H_2}$ (maps H_1 to H_2).

Lets take

$$H_I[h, J](\sigma) = \sum_i (h\sigma_i^x + J\sigma_i^z\sigma_{i+1}^z)$$

Consider the basic bonds

$$\{\sigma_i^z\sigma_{i+1}^z\}, \{\sigma_i^x\}$$

These two set of operators can generate a bond algebra because

- $(\sigma_i^z\sigma_{i+1}^z) = I$ and $(\sigma_i^x)^2 = I$
- Hermitian because Pauli Matrices
- Anti commutation relation between σ_i^x and $\sigma_{i-1}^z\sigma_i^z$ and $\sigma_i^z\sigma_{i+1}^z$ but commutation with everyone else.

- $\sigma_i^z \sigma_{i+1}^z$ anti commutes with σ_i^x and σ_{i+1}^x and commutes with everyone else. Thus we have by-commute.

From above we see that σ_i^x and $\sigma_i^z \sigma_{i+1}^z$ have symmetrical behavior.

3 The Toric Code (studied [5])

If I've slowly re-discovered anything again - if I don't understand something right away, I should wait until I've seen calculation related to what I didn't understand earlier.

Definition 5: Toric Code Hamiltonian

Consider 2D square lattice with periodic boundary condition (hence a torus) and each *edge* contains a spin 1/2 particle. Define the Hamiltonian

$$H = - \sum_v A_v - \sum_p B_p \quad (3.1)$$

where the first sum is over all vertices v and the second some is over all faces (or we'll call plaquette) p . The vertex operator A_v and plaquette operator B_p are *tensor products of Pauli operators* $X = \sigma^x$ and $Z = \sigma^z$ acting on individual spins.

$$A_v = \prod_{j \in \text{star}(v)} Z_j \quad \text{and} \quad B_p = \prod_{j \in \text{bdy}(p)} X_j$$

Here $\text{star}(v)$ is the set of edges adjacent to the vertex v and $\text{bdy}(p)$ is the set of edges forming the boundary of the plaquette p .

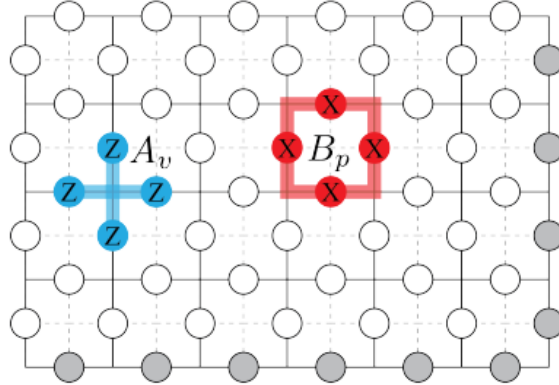


Figure 1. The Toric Code

On A_v and B_p commuting with each other: Note that for any overlap between A_v and B_p , the number of edges that overlap is always even. The X 's and Z 's always anti-commute, but because there's an even number of them, they cancel out.

Also, it's pointed out that each of the terms A_v nad B_p are Hermitian and square to 1. Hence the Eigenvalues are $+1, -1$.

Because the anti-commutation effect cancels out, the ground state (or states) will be simultaneous $+1$ eigenstate of all the A_v and B_p operators. **States that satisfy above constraints:** Each square can have custody of two spin sites (“left”, “bottom” for example, and the two other sites “right”, “top” belong to neighbors). Each site has a Hilbert space of dimension 2. For N unit cells, there are $2N$ sites, and thus total dimension being 2^{2N} .

Now, the paper requires us to think A_v and B_p as constraints on the Hilbert space, and violating any constraints costs energy so the ground state will obey every constraint. There’s a total of N number of A_v and B_p each. They each give $N - 1$ independent constraints because the product of all A_v is the identity, and similarly for B_p (it also has $N - 1$ constraints).

So in total we have $2N - 2$ constraints, so we expect a Ground State Manifold of dimension $2^{2N-(2N-2)} = 4$.

Ground State Construction

Notation

$$|0\rangle = |\uparrow\rangle \quad \text{and} \quad |1\rangle = |\downarrow\rangle$$

where we call $|1\rangle$ occupied and $|0\rangle$ as empty. So

$$X |0\rangle = |1\rangle \quad \text{and} \quad X |1\rangle = |0\rangle$$

Study of the occupied and empty states: Focus on A_v for v vertices. If a vertex v has an adjacent odd number of -1 occupied states, then the eigenvalue of the A_v for that vertex v is -1 .

Confusingly the note is saying that A_v *adjacent to an occupied state is* -1 . But I don’t see how because $A_v = Z_{v,\text{down}} Z_{v,\text{right}} Z_{v,\text{top}} Z_{v,\text{left}}$. How can anything outside of these 4 affect A_v ?

Closed loops of occupied sites can also end up giving $+1$ eigenvalue because of coming in pairs. I should remember this from Landau’s Statistical Mechanics solution of Ising model.

Now let’s introduce B_p . Because it’s made of X , so it takes the plaquette and flips the states. Just by some examples its obvious that B_p can create loops or deform existing loops. Another good observation is that B_p cannot break or join a string of occupied edges no matter what combination of B_p we use.

But here’s the **topological part**. A string of occupied states that is formed across the lattice (winds around the torus) cannot be contracted into nothing, unless it has been broken. This realization gives us 4 classes of loops which cannot be deformed into one another.

- Trivial Class with no loops.
- A loop that winds around the torus in either directions (hence two independent class is formed).
- Combination of two loops with one winding around in each direction.

4 Dummy Introduction

Theorem 2: Fluid Equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad \text{and} \quad \rho \frac{D\mathbf{u}}{Dt} = -\nabla P$$

and using Theorem [2](#)

References

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